

Minh Vu

Gettysburg, PA | vuminh300805@gmail.com | linkedin.com/in/minhvu | github.com/minhvdq | damianovu.com/

EDUCATION

Gettysburg College

Gettysburg, PA

Bachelor of Science in Computer Science and Mathematics

Dec 2026

- **Courses:** Data Structures and Algorithms, The Theory of Computation, Ethical Hacking, Computer Vision
- **Achievements:** Bronze Medal ICPC Mid-Atlantic Regional Contest 2024 at PSU-Harrisburg site

EXPERIENCE

CMC Global

Software Engineer Intern

Nov 2025 – Jan 2026

- Improved system modularity by engineering 7 microservice APIs for Payment Authorization and Courier Dispatch using **Java SpringBoot**, effectively decoupling Order and Delivery workflows into independent, scalable services
- Ensured data consistency across distributed transactions by implementing **Apache Kafka** and the **SAGA** pattern, enabling automated retry and rollback mechanisms for complex payment and delivery states
- Ensured UI state consistency by developing a **Vue.js** dashboard with **E2E testing**, utilizing **PostgreSQL** and the **Outbox pattern** to synchronize real-time order updates

Gettysburg College

Full-stack Developer

Jun 2025 – Jul 2025

- Developed and deployed a full-stack event management web app for the department using **React.js**, **Express.js**, and **AWS EC2** to handle event scheduling, attendance, and resource allocation
- Managed **200+** student-event registrations using **MySQL** junction tables with foreign key constraints to prevent duplicates and ensure data integrity
- Reduced event query response times by **80%** by implementing an **Amazon ElastiCache** layer to handle frequently accessed scheduling data

VinAI

Software Engineer Intern

Jun 2024 – Aug 2024

- Engineered a scalable distributed system for synchronized AI security camera clusters, enabling real-time object recognition and vector data propagation across multiple camera nodes to ensure global data consistency
- Architected a leader election system using **Apache ZooKeeper**, implementing watcher functions and sequential Z-nodes to automate master/slave failover; ensured **99.8%** uptime by instant re-vote based on the lowest-ID priority
- Streamlined camera-to-backend communication by developing **RESTful APIs** with **Express.js**, **MongoDB**, and orchestrating containerized deployments using **Docker** and **Kubernetes** to simulate multi-node environments

PROJECTS

AlphaGen | Python, Pytorch, Tensorflow, Keras, Hugging Face, FinBERT, OpenAI

- Architected an algorithmic trading microservices ecosystem integrating **PyTorch**, **TensorFlow**, **Hugging Face**, and **OpenAI**, processing **1000+** daily data points
- Engineered dual predictive data pipelines, achieving **92%** sentiment classification accuracy via a local **FinBERT** model and **85%** volatility forecasting accuracy via a **TensorFlow LSTM**
- Trained a **Deep Q-Network (DQN)** reinforcement learning agent in **PyTorch** to execute dynamic trades, outperforming a baseline buy-and-hold strategy by **14%** in historical backtesting
- Automated executive reporting via the **OpenAI API**, eliminating manual review time by synthesizing raw trading logs, sentiment scores, and volatility metrics into plain-English daily summaries

GoScale Mail | Go, Docker, Kubernetes, gRPC, Postgres, MongoDB, RabbitMQ, HTML, CSS

- Built a distributed, multi-microservice chat application in **Go**, featuring an email-style interface that supports sending the marketing message to multiple recipients for large-scale marketing campaigns
- Developed Auth service with **PostgreSQL** and integrated **OAuth2.0** to provide secure and seamless user sign-in
- Integrated **RabbitMQ** to orchestrate async message, auth and log requests, ensuring efficient backend flows
- Containerized and deployed services with **Docker** to enable independent scalability.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, Go, SQL, C++, HTML/CSS

Frameworks: React.js, Node.js, Express.js, Vue.js, SpringBoot, Docker, Kubernetes, Flask, Django

Tools: ZooKeeper, Kafka, RabbitMQ, Git, AWS, Amazon ElastiCache, Prometheus, Grafana, Postman, Figma, Vercel

Databases: MongoDB, PostgreSQL, MySQL, Redis, Firebase

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, Keras, FinBERT, OpenAI API, scikit-learn, NumPy